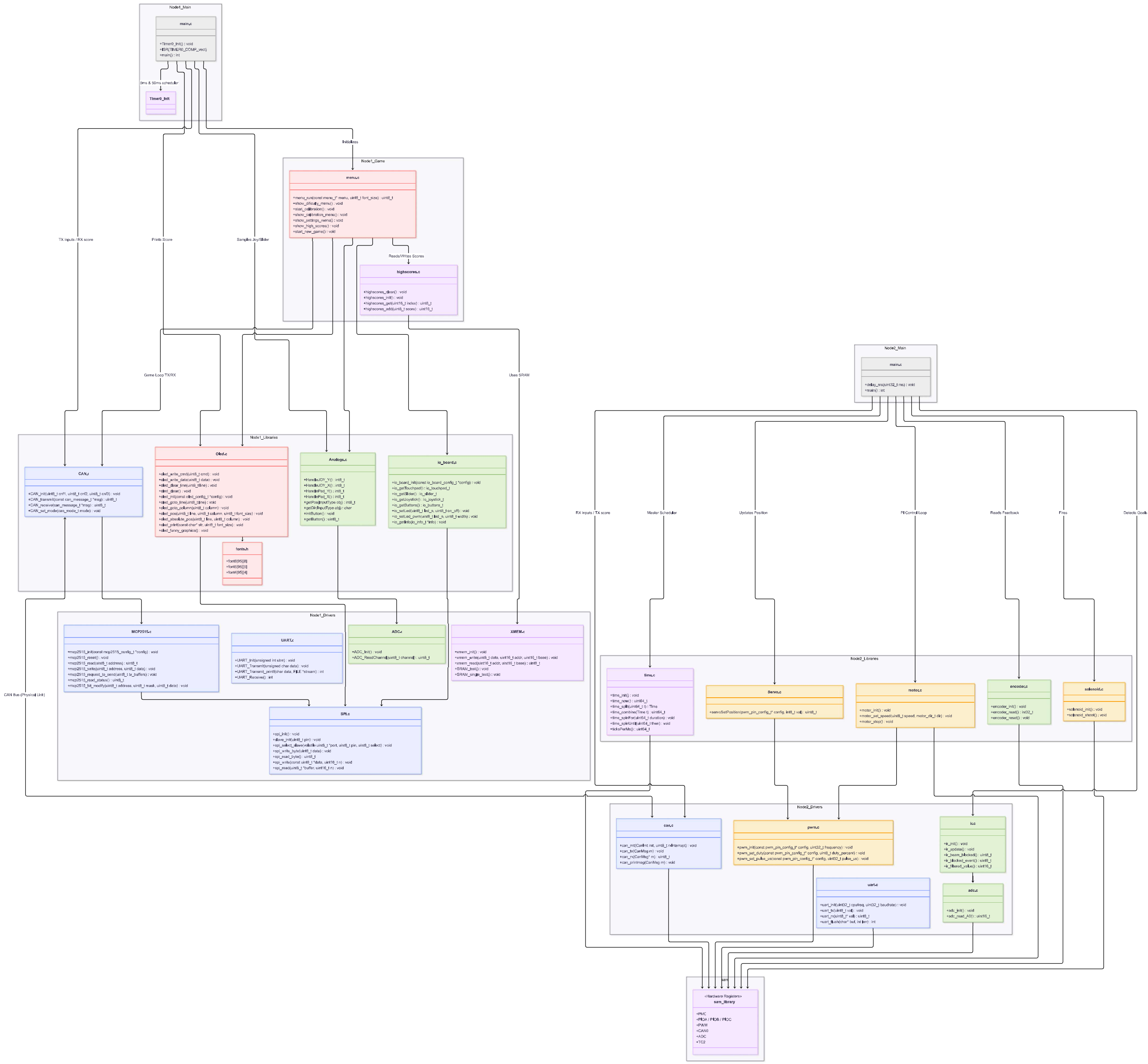
Node1_Main
main.c
+Timer0_Init() : void
+ISR(TIMER0_COMP_vect)
+main() : int
5ms & 50ms scheduler
Timer0_Init
Initializes
TX inputs / RX score
Prints Score
Samples Joy/Slider
Node1_Game
menu.c
+menu_run(const menu_t* menu, uint8_t font_size) : uint8_t
+show_dificulty_menu() : void
+start_calibration() : void
+show_calibration_menu() : void
+show_settings_menu() : void
+show_high_scores() : void
+start_new_game() : void
Reads/Writes Scores
highscores.c
+highscores_clear() : void
+highscores_init() : void
+highscores_get(uint16_t index) : uint8_t
+highscores_add(uint8_t score) : uint16_t
Game Loop TX/RX
Uses SRAM
Node1_Libraries
CAN.c
+CAN_init(uint8_t cnf1, uint8_t cnf2, uint8_t cnf3) : void
+CAN_transmit(const can_message_t *msg) : uint8_t
+CAN_receive(can_message_t *msg) : uint8_t
+CAN_set_mode(can_mode_t mode) : void
Oled.c
+oled_write_cmd(uint8_t cmd) : void
+oled_write_data(uint8_t data) : void
+oled_clear_line(uint8_t line) : void
+oled_clear() : void
+oled_init(const oled_config_t *config) : void
+oled_goto_line(uint8_t line) : void
+oled_goto_column(uint8_t column) : void
+oled_pos(uint8_t line, uint8_t column, uint8_t font_size) : void
+oled_absolute_pos(uint8_t line, uint8_t column) : void
+oled_print(const char* str, uint8_t font_size) : void
+oled_funny_graphics() : void
fonts.h
+font8[95][8]
+font5[95][5]
+font4[95][4]
Analogs.c
+HandleJOY_Y() : int8_t
+HandleJOY_X() : int8_t
+HandlePad_Y() : int8_t
+HandlePad_X() : int8_t
+getPos(InputType obj) : int8_t
+getDir(InputType obj) : char
+initButton() : void
+getButton() : uint8_t
io_board.c
+io_board_init(const io_board_config_t *config) : void
+io_getTouchpad() : io_touchpad_t
+io_getSlider() : io_slider_t
+io_getJoystick() : io_joystick_t
+io_getButtons() : io_buttons_t
+io_setLed(uint8_t led_n, uint8_t on_off) : void
+io_setLed_pwm(uint8_t led_n, uint8_t width) : void
+io_getInfo(io_info_t *info) : void
CAN Bus (Physical Link)
Node1_Drivers
MCP2515.c
+mcp2515_init(const mcp2515_config_t *config) : void
+mcp2515_reset() : void
+mcp2515_read(uint8_t address) : uint8_t
+mcp2515_write(uint8_t address, uint8_t data) : void
+mcp2515_request_to_send(uint8_t tx_buffers) : void
+mcp2515_read_status() : uint8_t
+mcp2515_bit_modify(uint8_t address, uint8_t mask, uint8_t data) : void
UART.c
+UART_Init(unsigned int ubrr) : void
+UART_Transmit(unsigned char data) : void
+UART_Transmit_printf(char data, FILE *stream) : int
+UART_Receive() : int
ADC.c
+ADC_Init() : void
+ADC_ReadChannel(uint8_t channel) : uint8_t
XMEM.c
+xmem_init() : void
+xmem_write(uint8_t data, uint16_t addr, uint16_t base) : void
+xmem_read(uint16_t addr, uint16_t base) : uint8_t
+SRAM_test() : void
+SRAM_single_test() : void
SPI.c
+spi_init() : void
+slave_init(uint8_t pin) : void
+spi_select_slave(volatile uint8_t *port, uint8_t pin, uint8_t select) : void
+spi_write_byte(uint8_t data) : void
+spi_read_byte() : uint8_t
+spi_write(const uint8_t *data, uint16_t n) : void
+spi_read(uint8_t *buffer, uint16_t n) : void
Node2_Main
main.c
+delay_ms(uint32_t ms) : void
+main() : int
RX inputs / TX score
Master Scheduler
Updates Position
PI Control Loop
Reads Feedback
Fires
Detects Goals
Node2_Libraries
time.c
+time_init() : void
+time_now() : uint64_t
+time_split(uint64_t t) : Time
+time_combine(Time t) : uint64_t
+time_spinFor(uint64_t duration) : void
+time_spinUntil(uint64_t then) : void
+ticksPerMs() : uint64_t
Servo.c
+servoSetPosition(pwm_pin_config_t* config, int8_t val) : uint8_t
motor.c
+motor_init() : void
+motor_set_speed(uint8_t speed, motor_dir_t dir) : void
+motor_stop() : void
encoder.c
+encoder_init() : void
+encoder_read() : int32_t
+encoder_reset() : void
solenoid.c
+solenoid_init() : void
+solenoid_shoot() : void
Node2_Drivers
can.c
+can_init(CanInit init, uint8_t rxInterrupt) : void
+can_tx(CanMsg m) : void
+can_rx(CanMsg* m) : uint8_t
+can_printmsg(CanMsg m) : void
pwm.c
+pwm_init(const pwm_pin_config_t* config, uint32_t frequency) : void
+pwm_set_duty(const pwm_pin_config_t* config, uint8_t duty_percent) : void
+pwm_set_pulse_us(const pwm_pin_config_t* config, uint32_t pulse_us) : void
ir.c
+ir_init() : void
+ir_update() : void
+ir_beam_blocked() : uint8_t
+ir_blocked_event() : uint8_t
+ir_filtered_value() : uint16_t
uart.c
+uart_init(uint32_t cpufreq, uint32_t baudrate) : void
+uart_tx(uint8_t val) : void
+uart_rx(uint8_t* val) : uint8_t
+uart_flush(char* buf, int len) : int
adc.c
+adc_init() : void
+adc_read_A0() : uint16_t
sam
«Hardware Registers»
sam_library
+PMC
+PIOA / PIOB / PIOC
+PWM
+CAN0
+ADC
+TC2